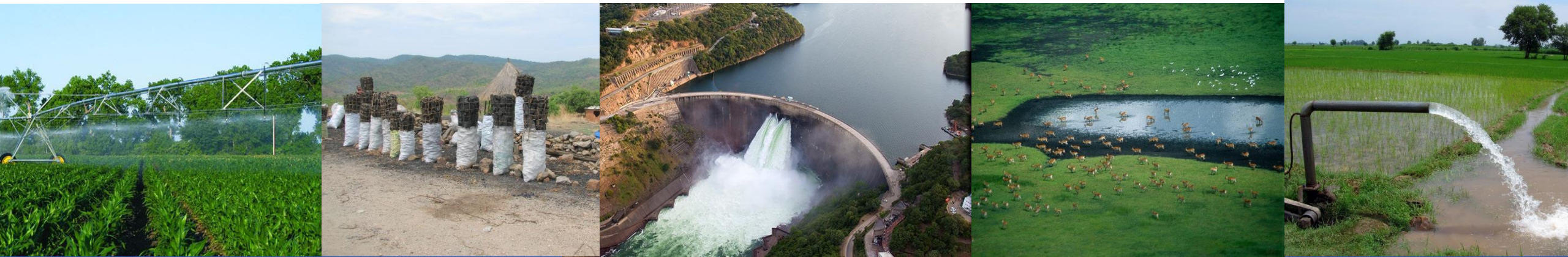


Crops in CWatM

Peter Burek, Carla Catania, Silvia Artuso
International Institute for Applied Systems Analysis
Research Scholars at
Water Research Group, BNR

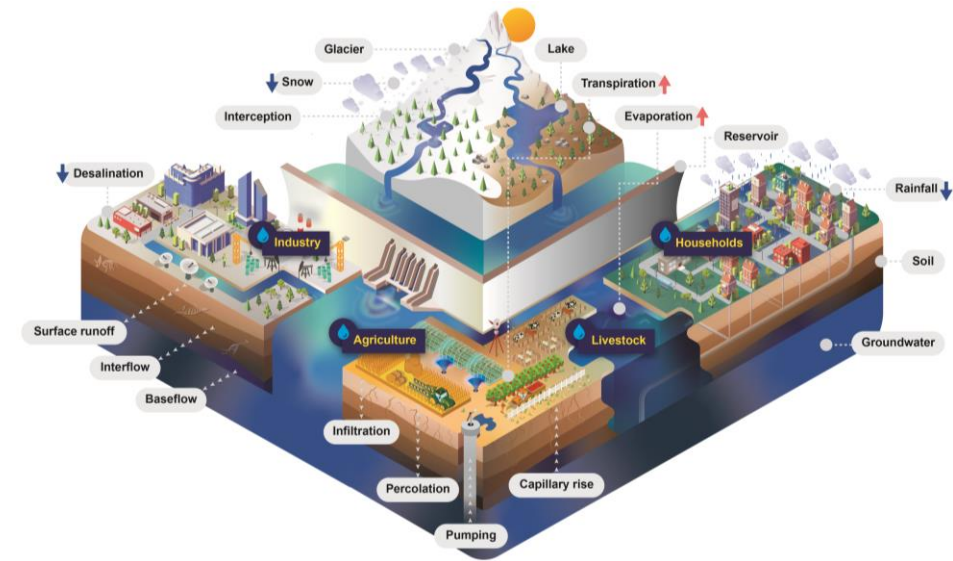


Why include crops in Community Water Model (CWatM)?

Agriculture is one of the specific processes related to human interactions within the water cycle that hydrologic models necessitate to include to simulate the water cycle in managed regions and to investigate questions of management, reliance, and security.

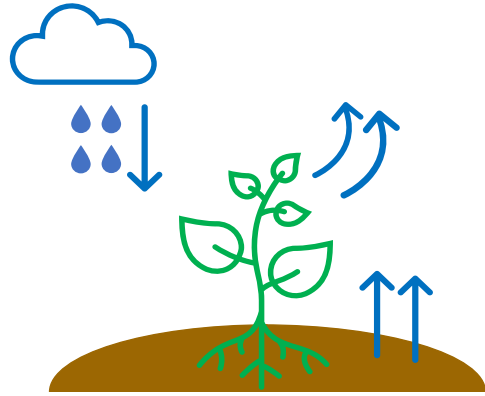
Refining Evapotranspiration (ET) in CWatM by including crops allows answering to questions like:

- What is the spatially-distributed crop-specific transpiration daily?
- How much irrigation water do specific crops use?
- What is the crop specific water deficit?



<http://www.iiasa.ac.at/cwatm>
<https://cwatm.iiasa.ac.at/>

Evapotranspiration is the transformation of liquid water into vapour through plant transpiration, soil moisture evaporation, and intercept evaporation.



CWatM calculates potential evapotranspiration, and actual evapotranspiration is limited by available soil moisture. Land cover distribution influences how water flows through and within cells.

Land classes in CWatM include:

forests



grasslands



irrigated
paddy



irrigated
land



sealed land



water



Land classes are input with raster map time-series (single value for each cell for each timestep) holding the associated land class fraction for each cell, updated annually; grasslands are allocated into the remaining land.

Crops are introduced into the grasslands and irrigated land classes.



How to include crops in Community Water Model (CWatM)?

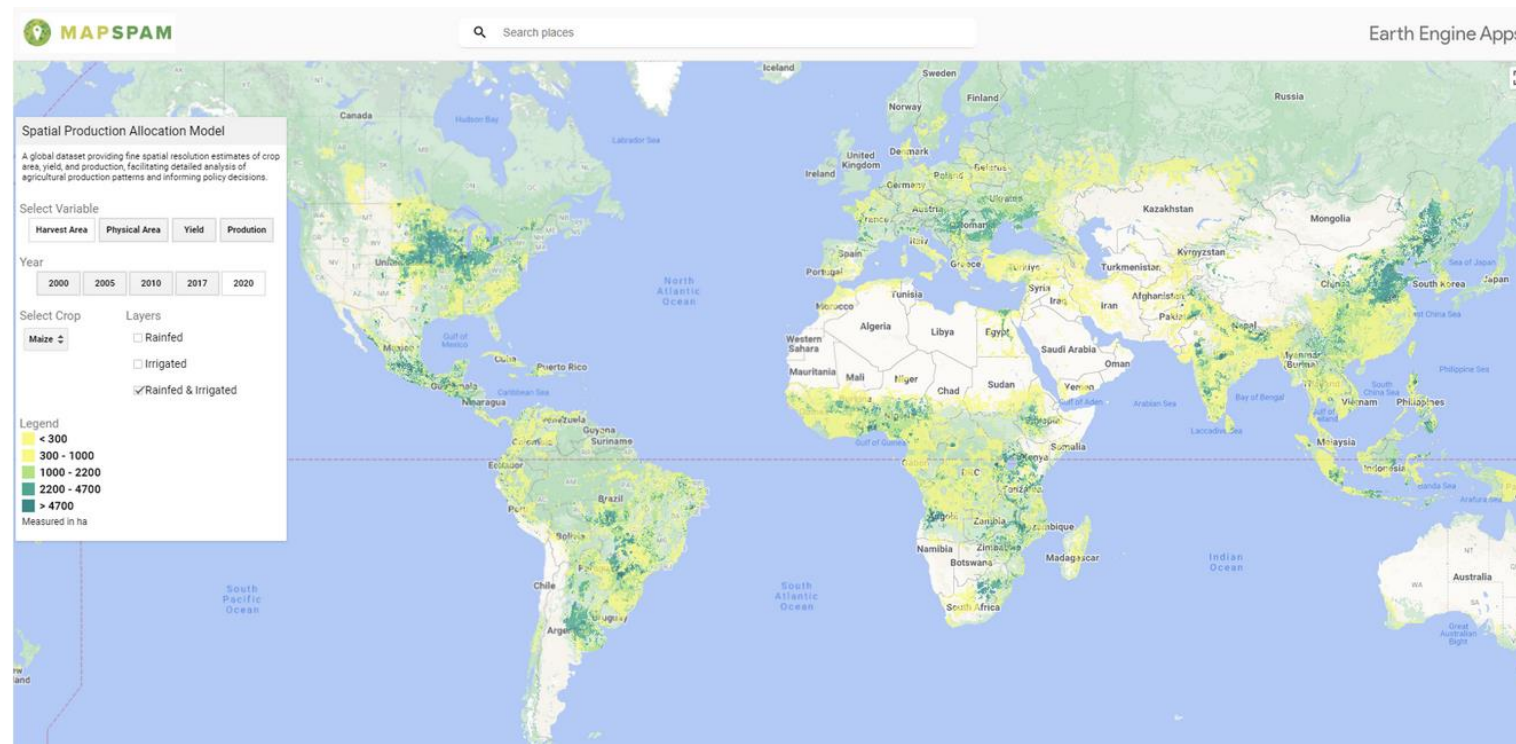
Crop-specific land and water use are included in CWatM through user-defined crops and parameters.

What will you need for running crops in CWatM:

1. Crop maps: raster maps holding the (potential) fractional crop areas separately for irrigated and non-irrigated.
2. Crop coefficients and crop parameters values for each crop → to be input in an excel file

Crop maps

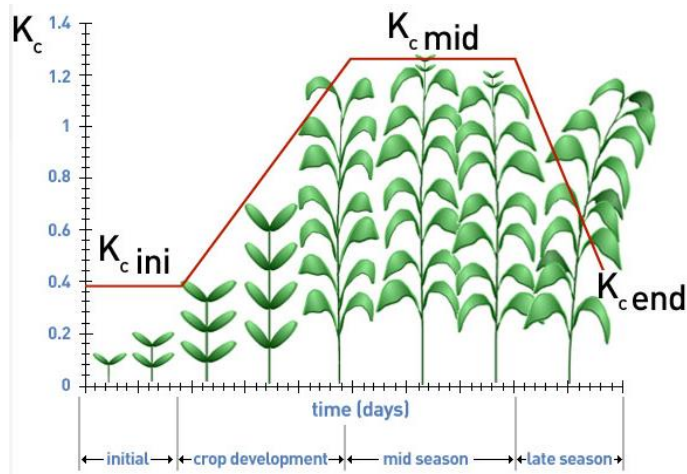
SPAM2020: global GEOTIFF maps (5 arc-minute grid cell) for 46 crops from IFPRI's Spatial Production Allocation Model (SPAM) for the year 2020 (SPAM2020 v1.0, International Food Policy Research Institute, 2024, <https://doi.org/10.7910/DVN/SWPENT>).



MIRCA2000: Global data set of 26 monthly irrigated and rainfed crop areas around the year 2000 with spatial resolution of 5 arc minutes
 GAEZ+ 2015: Monthly Cropland Data: Global gridded monthly crop physical area for 26 irrigated and rainfed crops at 5-minute resolution

Crop parameters

Crop choices and related parameters are user inputs through a spreadsheet. Crop parameters include planting month, length of four growth stages in months (GS), growth-stage specific crop coefficients (K_c) and yield sensitivities (K_y).



Crop characteristic	Stages of Development				
	Initial	Crop Development	Mid-season	Late	Total
Stage length, days	20	30	30	30	110
Depletion Coefficient, p	0.4	>>	0.5	0.65	-
Root Depth, m	0.25	>>	>>	0.8	-
Crop Coefficient, K_c	0.5	>>	1.1	0.8	-

<https://www.fao.org/land-water/databases-and-software/crop-information>

These parameters can be informed by crop- and region-specific studies, FAO Irrigation and drainage papers, FAO crop information, general studies referencing the crop-coefficient methodology, and GAEZ model documentation.

CWatM implementation

Crop water use is implemented in CWatM following the methodology of stage-specific crop coefficients (K_c) and yield sensitivities (K_y) from **FAO Irrigation and drainage papers** (Doorenbos et al. 1979; Allen and Pruitt 1991; Allen et al. 1998; Steduto et al. 2012).

Potential evapotranspiration for a crop can be determined by multiplying the reference evapotranspiration (estimation of the evapotranspiration from the "reference surface", i.e. a hypothetical grass reference crop with fixed parameters) by a **crop coefficient, K_c** , which is the ratio of potential crop evapotranspiration to the reference crop evapotranspiration.

$$PotET_{crop} = Kc_{crop} * ET_{ref}$$

Actual evapotranspiration is limited by soil moisture.

Crops are planted in the order they are provided in the spreadsheet.

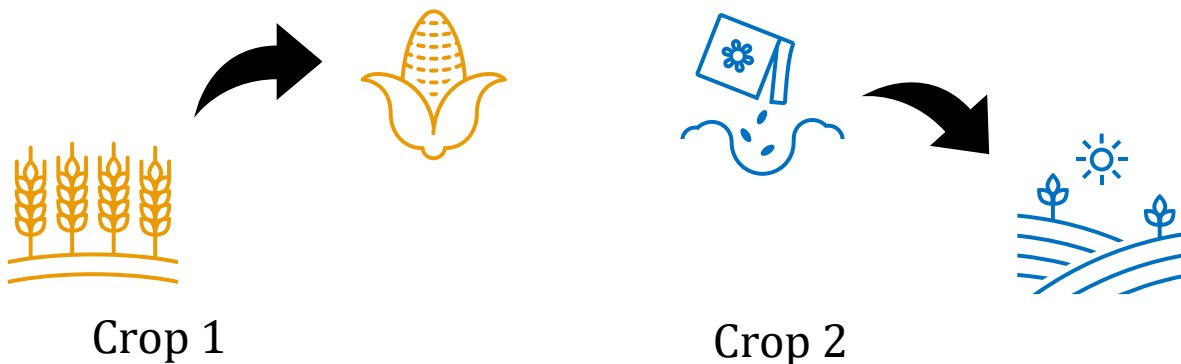
The timing of crop planting is as follows: on the first day of the month, crops with the final growth stage completed are harvested, and crops with the relevant planting month are sown, assuming sufficiently available land.

Crop type	Planting month	Harvest month
Crop 1	April	September
Crop 2	September	June
Crop 3	October	July

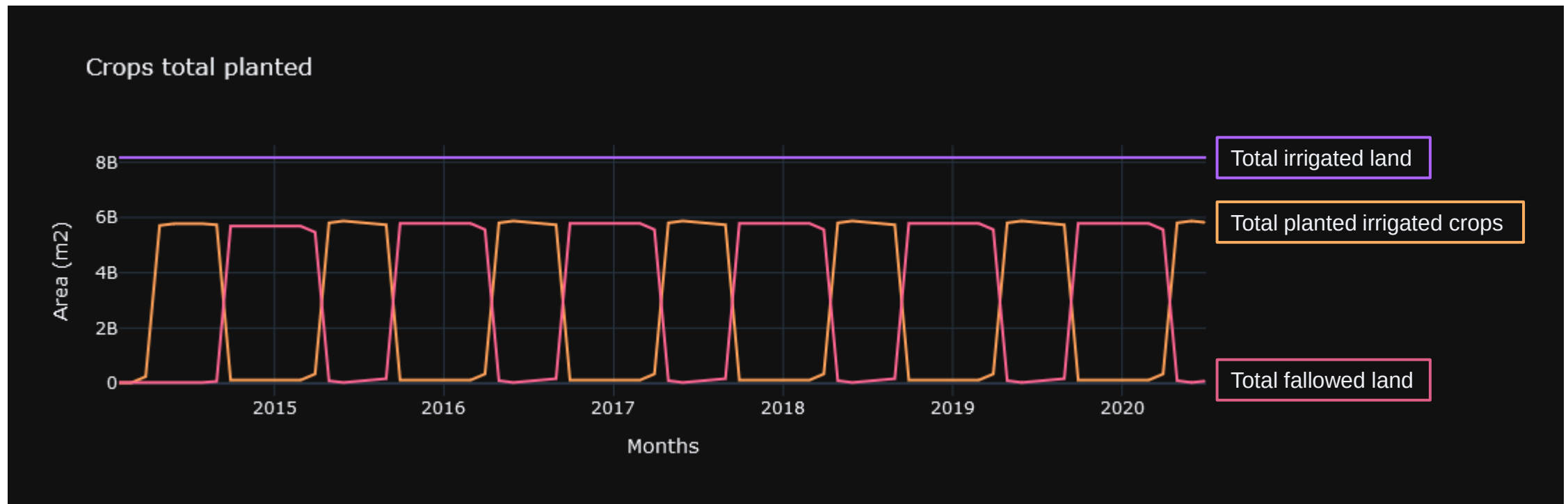
The amount of land available for irrigated and non-irrigated crops is given by the irrigated and grasslands land classes, respectively, minus the crops currently sown.

Paddy rice is treated in its own land class (irr_paddy).

Example: September



Fallow land is introduced and represents areas prepared for agriculture but not currently sown, namely, fields presently with only bare soil. Fallow land is determined automatically by replacing areas where crops are harvested and not immediately sown.



Irrigation practices representation

Irrigation is demanded for a cell with the irrigated land class only if there is more than bare soil; fallowed land does not receive irrigation. To determine when soil-moisture deficit instigates irrigation, a variable referred to as the **alpha depletion** α is introduced.

- $Moisture_{available} = Moisture_{current} - Moisture_{WP}$
- $Moisture_{available} < \alpha * Moisture_{critical} \rightarrow$
- $Irrigation_{demand} = \alpha * (Moisture_{FC} - Moisture_{WP}) - Moisture_{available}$

The alpha term facilitates representing irrigation practices that allow for drying to below critical moisture levels between irrigation applications and applying less than potential irrigation demand. Irrigation demand is further restricted by availability and the sector-, source- abstraction fractions, and is capped by infiltration capacity and scaled by inefficiency.

Data preprocessing: download GEOTIFF crop maps from SPAM2020

<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SWPENT>



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Global Spatially-Disaggregated Crop Production Statistics Data for 2020 Version 1.0

Version 3.0



International Food Policy Research Institute (IFPRI), 2024, "Global Spatially-Disaggregated Crop Production Statistics Data for 2020 Version 1.0", <https://doi.org/10.7910/DVN/SWPENT>, Harvard Dataverse, V3

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The 2020 SPAM (Spatial Production Allocation Model) products, encompassing crop area, yield, and

Data preprocessing: download GEOTIFF crop maps from SPAM2020

<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SWPENT>

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<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SWPENT>



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 SPAM harvested area, global pixels, files for 3 technologies



Citation Requirements

International Food Policy Research Institute (IFPRI). 2024. Global Spatially-Disaggregated Crop Production Statistics Data for 2020 Version 1.0.0. <https://doi.org/10.7910/DVN/SWPENT>. Harvard Dataverse. Version 3.

Disclaimer

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Name

Email *

Institution *

Position

Data preprocessing: download GEOTIFF crop maps from SPAM2020

spam2020V1r0_global_harvested_area.zip

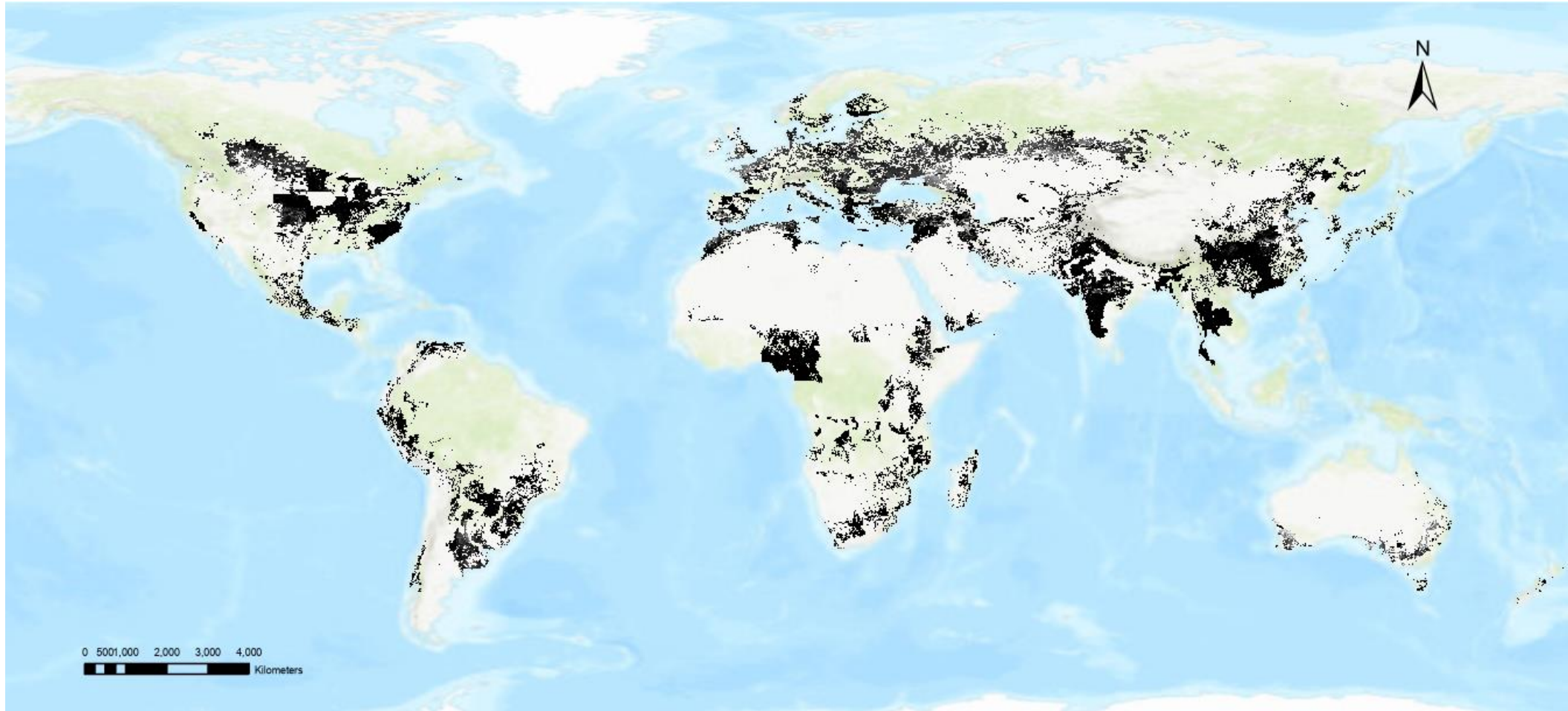
.. Danube_modelling > MapSPAM_2020 > spam2020V1r0_global_harvested_area

Sort View ...

Name	Date modified	Type	Size
spam2020_v1r0_global_H_BANA_I.tif.ovr	11/27/2024 11:18 AM	OVR File	94 KB
spam2020_v1r0_global_H_BANA_R.tif.ovr	5/13/2024 3:45 PM	OVR File	199 KB
spam2020_v1r0_global_H_BARL_I.tif.ovr	5/8/2024 10:39 AM	OVR File	109 KB
spam2020_v1r0_global_H_BARL_R.tif.ovr	5/13/2024 3:45 PM	OVR File	266 KB
spam2020_v1r0_global_H_BEAN_I.tif.ovr	5/8/2024 10:39 AM	OVR File	123 KB
spam2020_v1r0_global_H_BEAN_R.tif.ovr	5/13/2024 3:45 PM	OVR File	304 KB

Data preprocessing: crop maps from SPAM2020

WHEAT harvest area global map



Data preprocessing: obtain raster maps for the fractional area for each crop

Steps:

1. Clip the global raster maps to the basin area using the basin shape file
2. Translate from hectares to quadrat meters
3. Obtain the fraction of the crop area in each cell by dividing the crop area by the cell area
4. Resample the raster cell dimension from 5 arcminutes to 1 arcminutes

Maps need to be provided for both irrigated and non-irrigated crops. The two lists have to contain the same crops. In case some crops are represented only in one of the two regimes, an empty raster layer (with 0 values) needs to be provided as input for the missing map.

Data preprocessing: obtain raster maps for the fractional area for each crop

Steps:

1. Clip the global raster maps to the basin area using the basin shape file

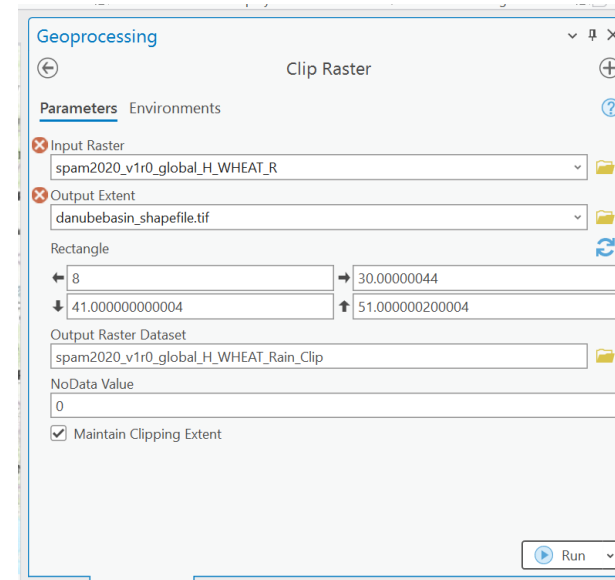
Function *Clip Raster*

Input Raster: raster to clip

Output Raster: raster containing the shape used for the clipping

Output Raster Dataset: clipped raster to save

Specify 0 as no data value and select “Maintain Clipping Extent”

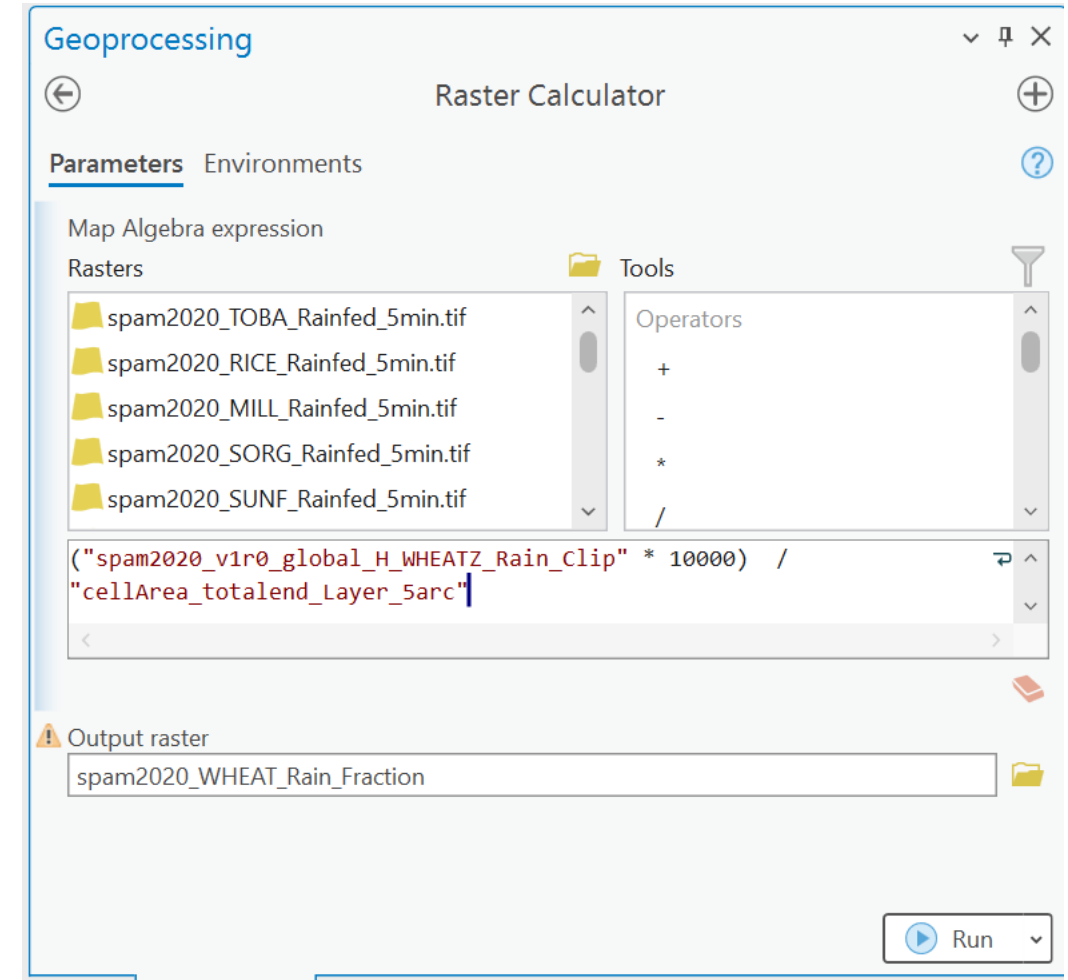


Data preprocessing: obtain raster maps for the fractional area for each crop

Steps:

2. Translate from hectares to quadrat meters (raster multiplied by 10000)
3. Obtain the fraction of the crop area in each cell by dividing the total crop area in quadrat meters by the cell area

Use *Raster Calculator*



Data preprocessing: obtain raster maps for the fractional area for each crop

Steps:

4. Resample the raster cell dimension from 5 arcminutes to 1 arcminutes

Use *Resample*

Geoprocessing

Resample

Parameters Environments

Input Raster

spam2020_WHEAT_Rain_Fraction

Output Raster Dataset

spam2020_WHEAT_Rainfed_1min

Output Cell Size

X

0.016666667

Y

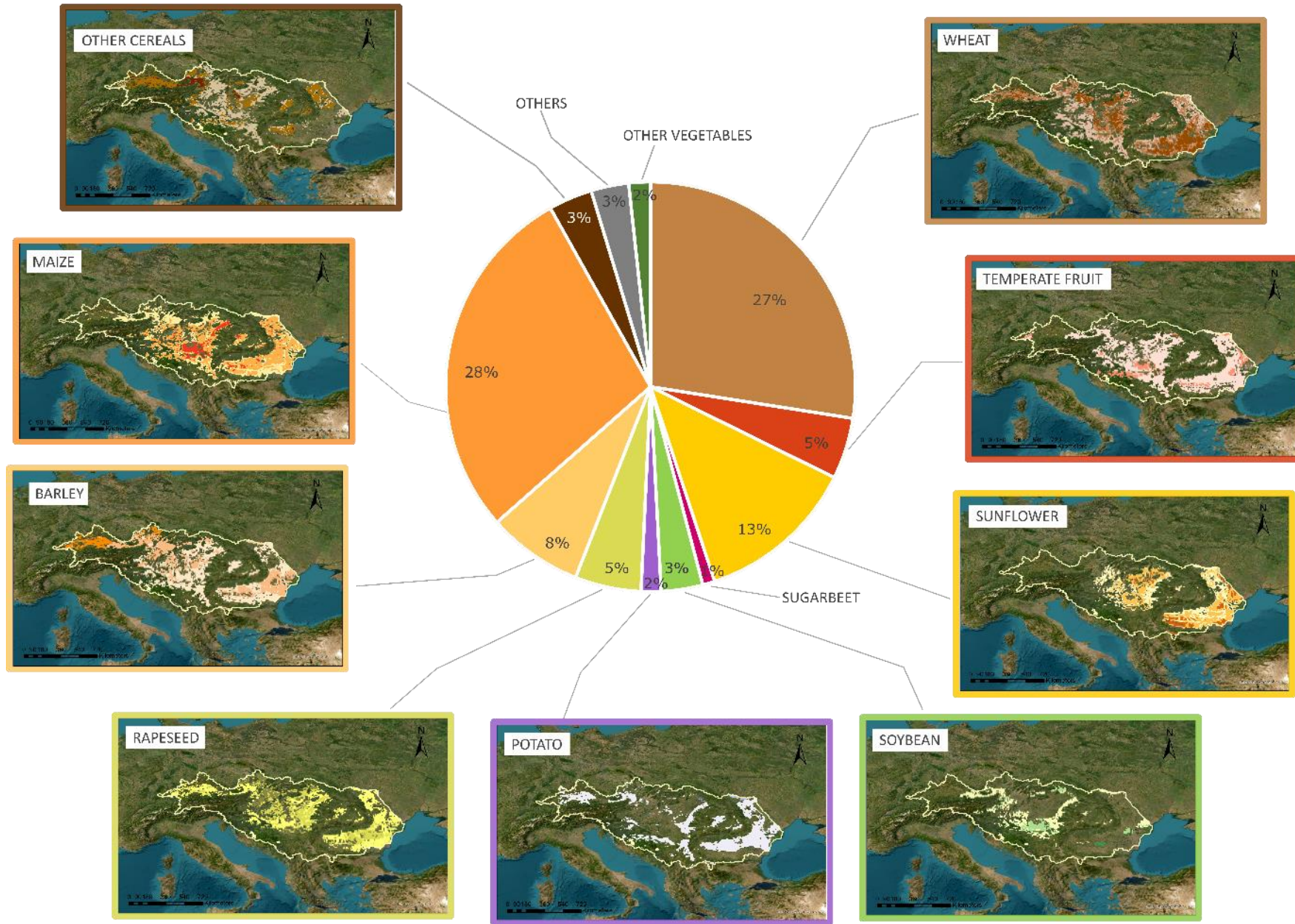
0.016666667

Resampling Technique

Nearest

Run

Crops in the Danube Basin



Data preprocessing: create excel settings file

Create an excel settings file containing the crop parameters. Those include planting month, length of four growth stages in months (GS), growth-stage specific crop coefficients (Kc) and yield sensitivities (Ky).

Table 1 List of crops included in the Danube basin modelling in CWatM, with the corresponding parameters.

Crop	Planting month	GS1	GS2	GS3	GS4	KC1	KC2	KC3	KC4	KY1	KY2	KY3	KY4
TOBACCO	5	1	1	1	1	0.5	0.8	1.1	0.8	1.00	1.00	1.00	1.00
WHEAT	9	2	5	1	1	0.4	0.775	1.15	0.25	1.00	1.00	1.00	1.00
<u>BARLEY Spring</u>	4	1	1	1	1	0.3	0.72	1.15	0.25	1.00	1.00	1.00	1.00
<u>BARLEY Winter</u>	9	2	3	3	2	0.3	0.72	1.15	0.25	1.00	1.00	1.00	1.00
BEAN	5	1	1	1	1	0.45	0.45	0.45	0.6	1.00	1.00	1.00	1.00
CHICKPEA	4	2	1	1	1	0.54	0.75	0.97	0.6	1.00	1.00	1.00	1.00
LENTIL	4	1	1	2	1	0.4	0.75	1.1	0.3	1.00	1.00	1.00	1.00
MAIZ	4	1	1	2	1	0.3	0.75	1.2	0.5	1.00	1.00	1.00	1.00
POTATO	4	1	1	2	1	0.5	0.8	1.15	0.5	1.00	1.00	1.00	1.00
SOYBEAN	4	1	1	2	1	0.5	0.8	1.15	0.5	1.00	1.00	1.00	1.00
SUGARBEET	3	1	2	2	1	0.35	0.77	1.2	0.85	1.00	1.00	1.00	1.00
<u>SUNFLOWER Irr</u>	4	1	1	2	1	0.35	0.75	1.15	0.35	1.00	1.00	1.00	1.00
<u>SUNFLOWER Rain</u>	4	1	1	2	1	0.35	0.67	1	0.35	1.00	1.00	1.00	1.00
SORGHUM	4	1	1	2	1	0.7	0.9	1	0.55	1.00	1.00	1.00	1.00
SMALL_MILLET	5	1	1	1	1	0.3	0.65	1	0.3	1.00	1.00	1.00	1.00

Some crops might have different Kc if they are irrigated or not/sowed in winter or summer. In these case, values need to be provided as separate entries in the table (and the corresponding maps).

Note: yield sensitivities (Ky) are not implemented yet. We can put 1 as a placeholder.

GS: growing stage; KC: Single crop coefficient (Kc); KY: yield sensitivities (Ky).

Data preprocessing: irrigated land map for the Danube basin

Table 2 Comparison of irrigated land (equipped and used) vs irrigated crops area in different datasets.

	Area equipped	Effectively used	Irrigated crop area
Standard 5min CWatM	36B m²	n/a	n/a
Meier	40.8B m²	n/a	n/a
GMIA	32.9B m²	5.3B m²	n/a
SPAM2020	n/a	n/a	8 B m²

New irrigated land map for the Danube basin

Created by summing of all irrigated crop maps in the Danube basin from SPAM2020 (with *Raster Calculator*)

How to input the crop in the CWatM settings file

- Option “includeCrops = True” needs to be input in [OPTIONS]
- Crop maps need to be input in two separate lists for irrigated and non-irrigated (rainfed) under [LANDCOVER].
- Excel settings file need to be provided under [FILE_PATHS], Excel_settings_file
- New irrigated map to be provided under [__irrNonPaddy], irrNonPaddy_fracVegCover, with option static_irrigation_map = True
- Update list of outputs needed (be careful with the crops numbering!).

Rice case

- Rice is a bit different. Maps of irrigated rice does not go with the other crops, but need to be input separately under `[__irrPaddy]`, `irrPaddy_fracVegCover`

```
[__irrPaddy]
PathIrrPaddy = $(FILE_PATHS:PathMaps)/landcover/irrPaddy

# Parameters for the Arno's scheme:
irrPaddy_arnoBeta = 0.2
# arnoBeta is defined by orographic,+ land cover add + calibration add, the soil water capacity distribution is based on this
# range [0.01 - 1.2]

# other paramater values

irrPaddy_minInterceptCap = 0.001
irrPaddy_cropDeplFactor = 0.0

irrPaddy_fracVegCover = P:\watmodel\CWATM\Regions\Danube_1min\Danube\input\landcover\crops\Irrigated_maps\spam2020_RICE_Irrigated_1min.tif
irrPaddy_rootFraction1 = $(PathIrrPaddy)/rootFraction1.map
irrPaddy_rootFraction2 = $(PathIrrPaddy)/rootFraction2.map
irrPaddy_maxRootDepth = $(PathIrrPaddy)/maxRootDepth.map
irrPaddy_minSoilDepthFrac = $(PathIrrPaddy)/minSoilDepthFrac.map
```

Analyze the outputs: Land toolbox

GitHub\CWatM\Toolkit\Notebooks\Land.ipynb

 .ipynb_checkpoints	12/5/2024 3:30 PM	File folder	
 functions	12/5/2024 4:23 PM	File folder	
 Outputs	5/6/2024 2:07 PM	File folder	
 ~ Waves.ipynb	5/6/2024 2:07 PM	IPYNB File	4,023 KB
 ○ Watercycles.ipynb	8/26/2024 11:25 AM	IPYNB File	8,877 KB
 ○ Watercycles-Copy3.ipynb	5/6/2024 2:07 PM	IPYNB File	23,987 KB
 ○ Watercycles-Copy4.ipynb	5/6/2024 2:07 PM	IPYNB File	9,951 KB
 Untitled.ipynb	11/28/2024 10:49 AM	IPYNB File	1 KB
 🌱 Land.ipynb	12/5/2024 6:36 PM	IPYNB File	9,148 KB
 📊 Excel outputs from NetCDF.ipynb	5/6/2024 2:07 PM	IPYNB File	8 KB

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Land toolbox

GitHub\CWatM\Toolkit\Notebooks\Land.ipynb

jupyter
Land Last Checkpoint: 5 hours ago

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

Land

Distributions of different land types and crops as well as select water uses.

- ✓ Update the file paths below.
- ▶ Run with the black and white fast-forward icon above.

```
[5]: output_folder = r'P:/watmodel/CWATM/Regions/Danube_1min/Danube/Outputs/out_SA_newIrrmap'
xl_settings_file_path = r'P:/watmodel/CWATM/Regions/Danube_1min/Danube/input/landcover/crops2/Crop_List_Danube_MAPSPAM2020.xlsx'
withCrops = True
withYield=False
```

Flow through different land types

The six specific land covers include forest, grasslands/rainfed agriculture, non-flood irrigation, flood irrigation, sealed land, and waterbody. Flow of water through and within cells is influenced by the distribution of land types. Water cycle flows are addressed separately for each land type. Agricultural lands includes different crops, as well as general representative crops and fallow/empty land.

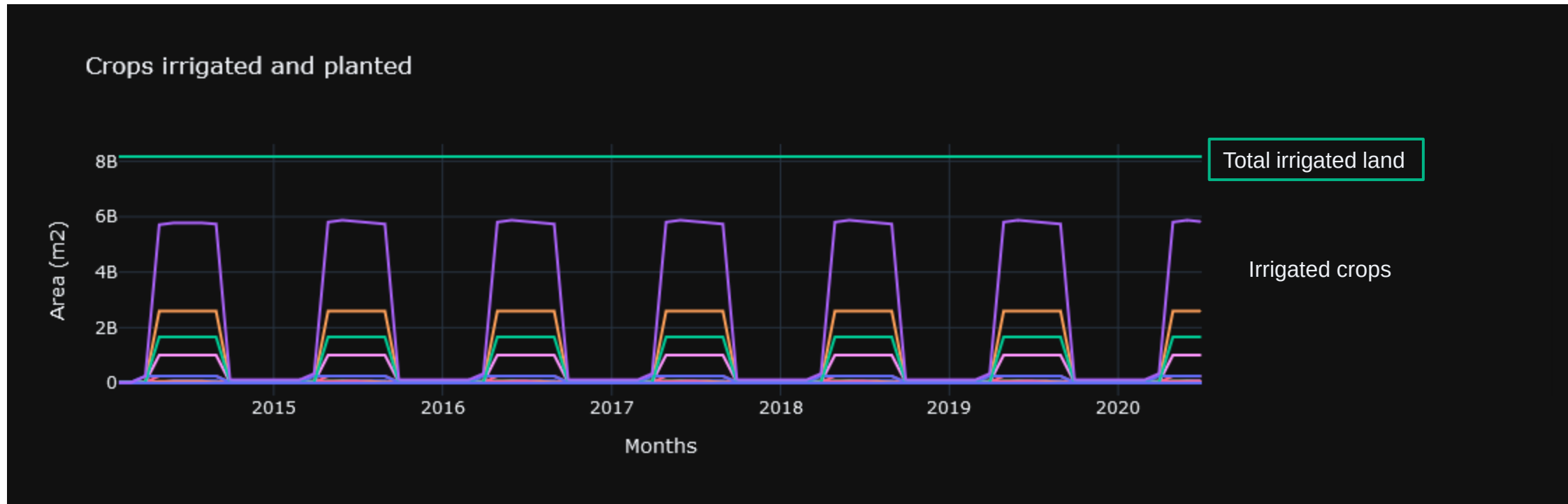
```
[2]: run functions/functions_Maps.ipynb
```

```
[3]: display_map('fracVegCover[3]_monthend', day=-1, Rad=8, colourScale = 'BrBG', dark=True,outputFolder = output_folder) #)2.6) #colourScale='balance_r'
```

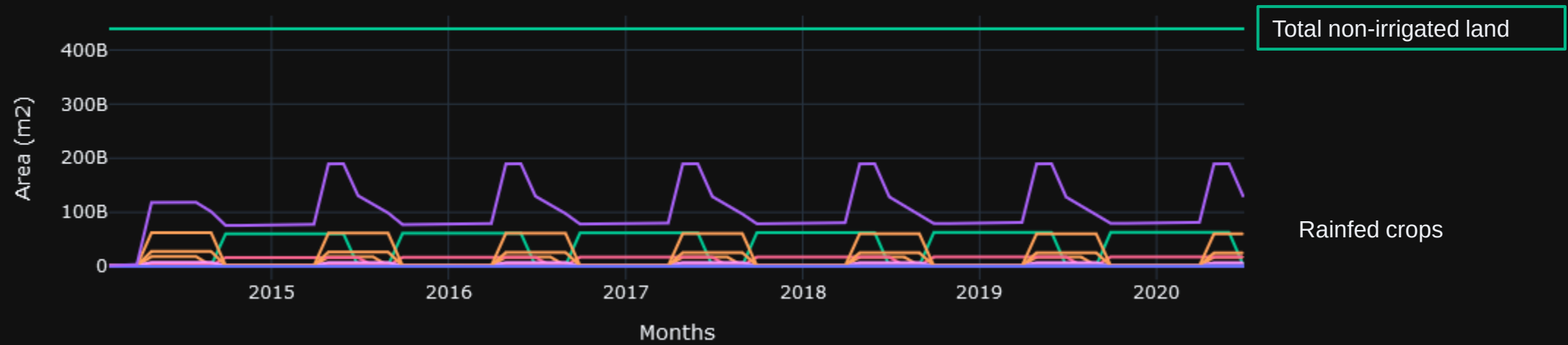
fracVegCover[3]_monthend



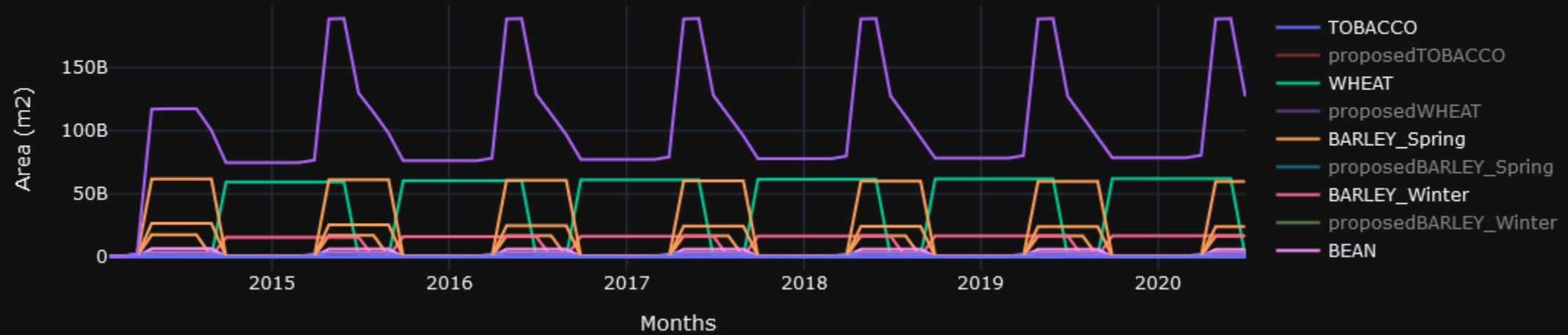
Crop outputs in CWatM for the whole Danube



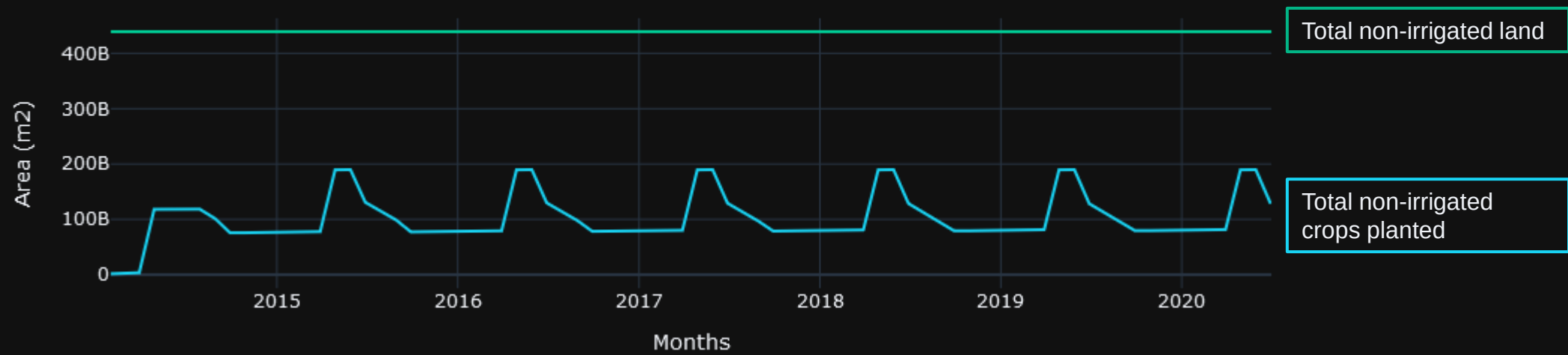
Crops non-irrigated and planted



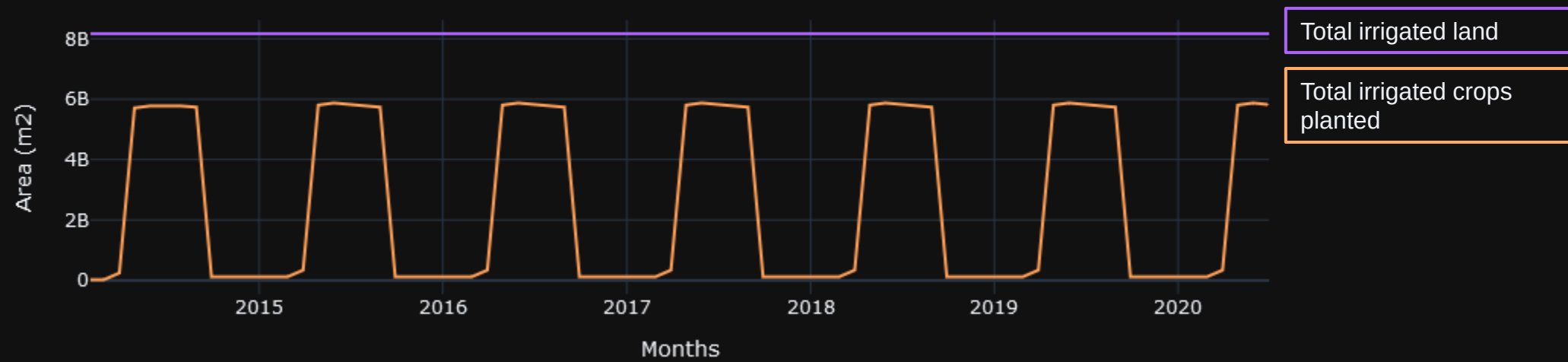
Crops non-irrigated and planted



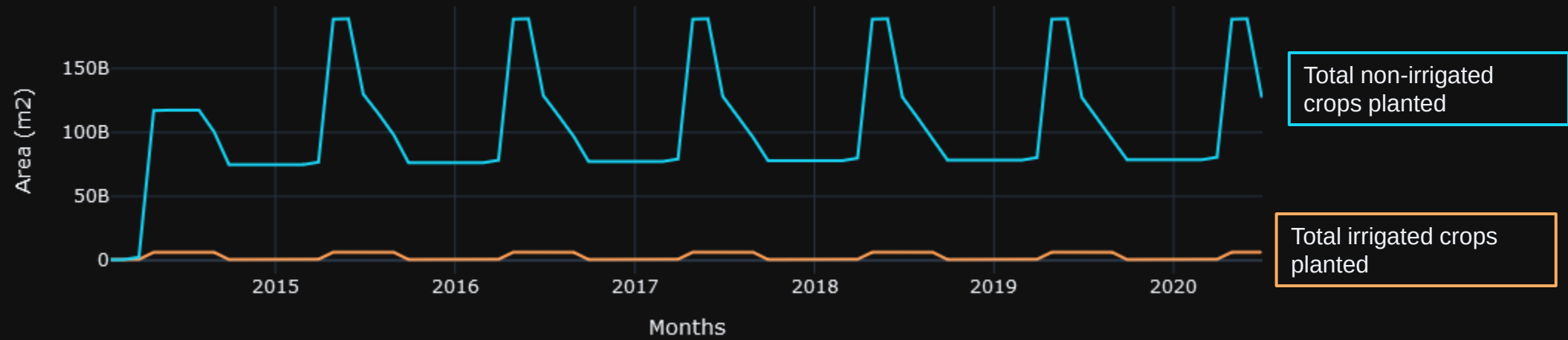
Crops total planted



Crops total planted

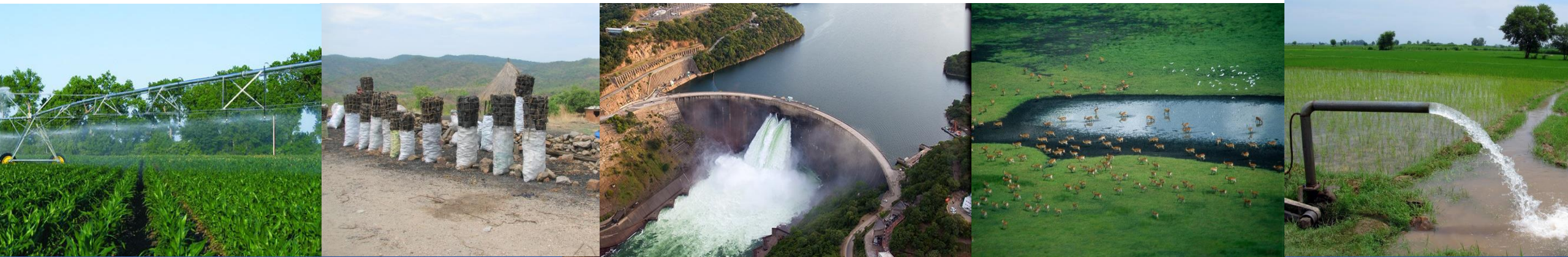


Crops total planted



Crops in CWatM: tutorial

Peter Burek, Carla Catania, Silvia Artuso
International Institute for Applied Systems Analysis
Research Scholars at
Water Research Group, BNR



Input crop maps in CWatM

In the “Exercise5_crops” folder, you will find some already preprocessed .TIFF crop maps for the whole Danube basin in “Irrigated_maps” and “Rainfed_maps”:

- Wheat
- Mais
- Potato
- Rice

In the same folder you also find FAO crops info sheets, excel settings file and CWatM settings file.

Tasks:

1. Update excel file with the crop parameters
2. Update the CWatM settings file:
 - a) Insert crop maps
 - b) Insert irrigated rice map. *Be careful, the rice irrigated map needs to go under irrigated paddy!*
 - c) Update outputs list (next slide)

Input crop maps in CWatM

Tasks:

1. Update excel file with the crop parameters

Fill put missing information in the file Crop_Parameters_Danube.xlsx, by using the FAO crop info sheets.

Crop	Planting mo	GS1	GS2	GS3	GS4	KC1	KC2	KC3	KC4	KY1	KY2	KY3	KY4
WHEAT	9	2	5	1	1	0.4	0.775		0.25	1.00	1.00	1.00	1.00
MAIZ	4		1	2	1		0.75	1.2		1.00	1.00	1.00	1.00
POTATO	4	1	1	2		0.5	0.8	1.15	0.5	1.00	1.00	1.00	1.00
RICE	4	1	1	2	1	1.05	1.12	1.2	0.75	1.00	1.00	1.00	1.00

Input crop maps in CWatM

Update the CWatM settings file:

a) Insert crop maps

```
#-----
[LANDCOVER]
PathLandcover = $(FILE_PATHS:PathMaps)/landsurface

coverTypes = forest, grassland, irrPaddy, irrNonPaddy, sealed, water
coverTypesShort = f, g, i, n, s, w
fractionLandcover = $(PathLandcover)/fractionLandcover_lisf.nc

# Landcover can vary from year to year
dynamicLandcover = True
# if landcover cannot vary, which year should be taken as fixed year
fixLandcoverYear = 1990

WHEAT_Irr = 
MAIZ_Irr = 
POTATO_Irr = 
RICE_Irr = P:\watmodel\CWATM\Regions\Danube_1min_SP\Tutorial\Exercise_crops\Irrigated_maps\spam2020_ZeroLayer_1min.tif

WHEAT_nonIrr = 
MAIZ_nonIrr = 
POTATO_nonIrr = 
RICE_nonIrr = 
```

Input crop maps in CWatM

Update the CWatM settings file:

- a) Insert crop maps
- b) Insert irrigated rice map. ***Be careful, the rice irrigated map needs to go under irrigated paddy!***

```
[__irrPaddy]
PathIrrPaddy = $(FILE_PATHS:PathMaps)/landcover/irrPaddy

# Parameters for the Arno's scheme:
irrPaddy_arnoBeta = 0.2
# arnoBeta is defined by orographic,+ land cover add + calibration add, the soil water capacity distribution is based on this
# range [0.01 - 1.2]

# other parameter values

irrPaddy_minInterceptCap = 0.001
irrPaddy_cropDeplFactor = 0.0

irrPaddy_fracVegCover = 
irrPaddy_rootFraction1 = $(PathIrrPaddy)/rootFraction1.map
irrPaddy_rootFraction2 = $(PathIrrPaddy)/rootFraction2.map
irrPaddy_maxRootDepth = $(PathIrrPaddy)/maxRootDepth.map
irrPaddy_minSoilDepthFrac = $(PathIrrPaddy)/minSoilDepthFrac.map
```

Input crop maps in CWatM

Update Outputs list:

OUT_MAP_Daily: irr_crop[0], irr_crop[1], irr_crop[2], irr_crop[3], actTransTotal_crops_nonIrr[0], actTransTotal_crops_nonIrr[1], actTransTotal_crops_nonIrr[2], actTransTotal_crops_nonIrr[3], ET_crop_nonIrr[0], ET_crop_Irr[0], ET_crop_nonIrr[1], ET_crop_Irr[1], ET_crop_nonIrr[2], ET_crop_Irr[2], ET_crop_nonIrr[3], ET_crop_Irr[3], ETRef, totalET, actualET[0], actualET[1], actualET[2], actualET[3], actualET[4]

OUT_MAP_MonthEnd: fracVegCover[0], fracVegCover[1], fracVegCover[2], fracVegCover[3], fracVegCover[4], fracVegCover[5], frac_totalIrr, frac_totalnonIrr, availableArableLand, fallowIrr, fallownonIrr, GeneralCrop_Irr, GeneralCrop_nonIrr, fracCrops_Irr[0], fracCrops_Irr[1], fracCrops_Irr[2], fracCrops_Irr[3], fracCrops_nonIrr[0], fracCrops_nonIrr[1], fracCrops_nonIrr[2], fracCrops_nonIrr[3], fracCrops_IrrLandDemand[0], fracCrops_IrrLandDemand[1], fracCrops_IrrLandDemand[2], fracCrops_IrrLandDemand[3], fracCrops_nonIrrLandDemand[0], fracCrops_nonIrrLandDemand[1], fracCrops_nonIrrLandDemand[2], fracCrops_nonIrrLandDemand[3]

Optional: Visualize crop outputs in CWatM

In the folder ...\\Tutorial\\Exercise5_crops\\Outputs you will find results from the Exercise5_crops CWatM run.

You can open the land notebook in the CWatM Toolkit folder with Jupyter Notebook and explore the results.

GitHub\\CWatM\\Toolkit\\Notebooks\\Land.ipynb

📁 .ipynb_checkpoints	12/5/2024 3:30 PM	File folder	
📁 functions	12/5/2024 4:23 PM	File folder	
📁 Outputs	5/6/2024 2:07 PM	File folder	
📄 ~ Waves.ipynb	5/6/2024 2:07 PM	IPYNB File	4,023 KB
📄 ○ Watercycles.ipynb	8/26/2024 11:25 AM	IPYNB File	8,877 KB
📄 ○ Watercycles-Copy3.ipynb	5/6/2024 2:07 PM	IPYNB File	23,987 KB
📄 ○ Watercycles-Copy4.ipynb	5/6/2024 2:07 PM	IPYNB File	9,951 KB
📄 Untitled.ipynb	11/28/2024 10:49 AM	IPYNB File	1 KB
📄 🌿 Land.ipynb	12/5/2024 6:36 PM	IPYNB File	9,148 KB
📄 📊 Excel outputs from NetCDF.ipynb	5/6/2024 2:07 PM	IPYNB File	8 KB

Open with Jupyter Notebook